MARK SCHEME · TEACHER COPY

11BR · Topic 1: Information Representation

Cambridge AS Level Computer Science 9618

Paper 1 Theory Fundamentals

Total marks: 75

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

For teacher / admin use only. Do not give to students before they have attempted the paper.

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Question 1 · May/June 2024 Paper 13, original Q1 (a),(b),(c),(d) [9 marks]

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>1(a)</td><td>1 mark for each correct answer:

• binary
• 3072
• denary/decimal
• 2000

A kibibyte has a binary prefix. Three kibibytes is the same as 3072 bytes.
A megabyte has a decimal/denary prefix. Two terabytes is the same as 2000 gigabytes.</td><td>4</td></tr>
<tr><td>1(b)</td><td>1 mark for correct answer:
F1</td><td>1</td></tr>
<tr><td>1(c)</td><td>1 mark for a correct answer:

The answer is too long to be represented in the same number of bits as the binary numbers being added</td><td>1</td></tr>
<tr><td>1(d)(i)</td><td>1 mark for all 3 answers correct:
<table><tr><th>Character set</th><th>Number of bits</th></tr><tr><td>ASCII</td><td>7</td></tr><tr><td>extended ASCII</td><td>8</td></tr><tr><td>Unicode</td><td>16/32</td></tr></table></td><td>1</td></tr>
</table>

1(d)(ii)	**1 mark** each: • Each character has a **unique** binary code • The binary code for each character is stored in sequence	**2**

Question 2 · May/June 2024 Paper 13, original Q2 (a),(b),(c) [9 marks]

Question	Answer	Marks
2(a)	**1 mark** for working: • 4000 * 3000 * 4 **1 mark** for correct answer: • 48MB	**2**
2(b)(i)	**1 mark** each to **max 3**: • The file takes less storage space on the web server than if lossless compression was used • The file is faster to upload/download to/from the server than if lossless compression was used • The file uses less bandwidth to transmit than if lossless compression was used • The file consumes less data allowance than if lossless compression was used	**3**
2(b)(ii)	**1 mark** each: • Identifies **consecutive** repeating **pixels** of the same colour • Stores the colour /pattern and the number of times it repeats	**2**

2(c)	**1 mark** each to **max 2**: • Colour/bit depth • Image resolution	**2**

Question 3 · May/June 2025 Paper 12, original Q2 (a),(b),(c) [7 marks]

Question	Answer	Marks
3(a)	**1 mark** for: 0010 0010 1110 **1 mark** for: 22E	**2**
3(b)(i)	**1 mark** for: -30	**1**
3(b)(ii)	**1 mark** for smallest, **1 mark** for largest Smallest: 1000 0000 Largest: 0111 1111	**2**

3(c)	**1 mark** for application and **1 mark** for corresponding justification e.g. • an application that performs financial / banking calculations • because financial transactions use only two decimal places and must be accurate, no accumulating / rounding errors and it is difficult to represent decimal values exactly in normal binary. Or • electronic displays • because visual displays only need to show individual digits • because conversion between denary and BCD is straightforward Or • The storage of the date and time in the BIOS of a PC • because conversion between denary and BCD is more straightforward Or • Barcode systems • because conversion between denary and BCD can be accurately completed	**2**

Question 4 · May/June 2026 Paper 12, original Q1 (a),(b),(c),(d) [8 marks]

Question	Answer	Marks	Guidance
4	Examiners must award marks for each response in line with: • the specific **content** defined in the mark scheme • the specific **skills** defined in the mark scheme or in the level descriptions • the **standard of response** required as exemplified by the standardisation scripts.		
4(a)	**1 mark** for a correct tick • The smallest element of an image that can be one colour	**1**	Correct answer only

4(b)	**1 mark** for each correct answer • Image resolution is the number of **pixels wide** by **pixels high** that form the image and screen resolution is the number of **pixels wide** by **pixels high** that can be **displayed** on a screen. • If screen resolution is not as high as image resolution the quality of the image is worsened. // Image resolution can be changed, but screen resolution has a physical limit	**2**	Accept horizontal and/or vertical for wide and/or high Just number of pixels for either or both is NE Do not accept bits or characters for pixels
4(c)	**1 mark** for a correct answer For example: • File size • File type • Dimensions of the image // image size • Compression type • Data offset – bytes from the beginning of the file to the actual pixel data	**1**	Mark the first answer given Do not accept resolution or colour depth, they are given in the question Do not accept a description of a file header or metadata Accept height and/or width for dimensions BOD
4(d)(i)	**1 mark** for: • Two // 2	**1**	Correct answer only
4(d)(ii)	**1 mark** for each correct point to **max 2** • Run Length Encoding (RLE) can be used • A series of **consecutive** pixels of the **same colour //** a **sequence** of the **same colour** pixel is stored as a single copy of the colour and the number of **consecutive** occurrences **1 mark** for a suitable reference **to the given image** • RLE: e.g. R2 // W4 // 2R4W // R2W4	**3**	Take care that the answers are compressing pixels not anything else Do not accept bits or characters or elements etc. for pixels If no reference to the diagram included in the answer, please check the diagram for annotations

4	0		0	0
4	1		0	1

Question 5 · May/June 2026 Paper 13, original Q3 (a),(b),(c) [13 marks]

Question	Answer	Marks	Guidance
5	Examiners must award marks **positively**. This means that: • marks are not deducted for errors or omissions • marks are awarded for correct/valid answers as defined in the mark scheme and also for **other valid answers** which go beyond the scope of the syllabus and mark scheme referring to your team leader as appropriate • the quality of spelling, punctuation and grammar are judged **only** when the mark scheme indicates that these features are **specifically assessed** in the question. However, the meaning of all responses must be unambiguous.		
5(a)	**1 mark** for: **Each sample** is stored as a 16-bit binary number // there are 2^16 / 65536 **different** amplitudes that can be recorded	**1**	Accept 2 bytes Or any equivalent.
5(b)(i)	**1 mark** each to **max 3** • Image is a series of pixels • … where each pixel has one colour • Each colour has a **unique** binary number • The binary number/code of the colour of each pixel is stored in sequence	**3**	
5(b)(ii)	**1 mark** for working **1 mark** for answer • 2048 * 1536 * 2 * 8 / 1024 / 1024 • 48 MiB	**2**	
5(c)(i)	**1 mark** each **max 2** for image and **max 2** for sound Image e.g. • Run-length encoding / RLE • Store each colour and the number of times it appears **consecutively** Sound e.g. • Reduce amplitude values (dynamic range) to only those sounds used in the actual recording … • … reducing sampling resolution without impacting the sound • Record changes in amplitudes instead of each individual amplitude … • …each change requires less bits than a new amplitude	**4**	Accept RLE for sound but do not allow for both parts

5(c)(ii)	**1 mark** for: • **Storage** located on a remote server that **anyone** can sign up to and use	**1**	Not commercial sites
5(c)(iii)	**1 mark** each to **max 2** • Animator can access anywhere that has internet access // animator can work on the animation in different locations as long as they have internet • Animator can access from any device // animator can work on different devices • Animator can give authentication/access to selected people without having to transfer the files/animation individually // Online collaboration with colleagues • Storing data in the cloud reduces the need for large amounts of internal storage on the local devices • The cloud company manages backups etc. • The Animator not responsible for security • More storage capacity readily available if needed	**2**	

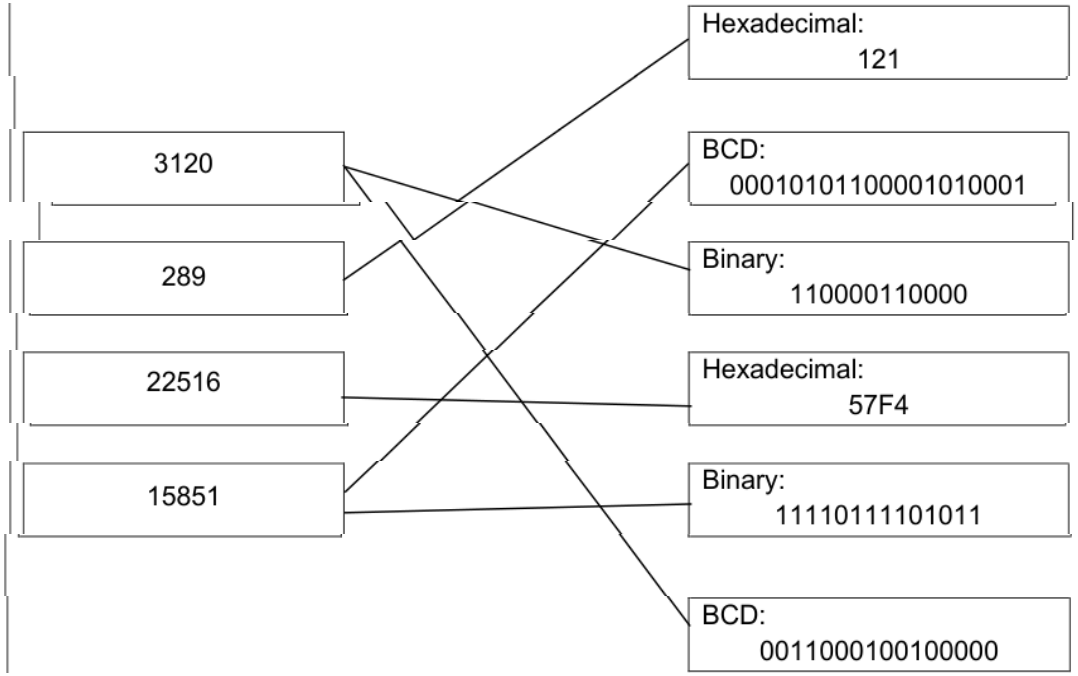
Question 6 · May/June 2026 Paper 13, original Q6 (a),(b),(c),(d) [8 marks]

Question	Answer	Marks	Guidance
6	Examiners must award marks according to the mark scheme and **must not** award marks with grade thresholds or grade descriptions in mind.		

Annotation	Meaning
	Correct
	Incorrect
	To indicate where a key word/phrase/code is missing.
	Not relevant or used to separate parts of an answer.
	Indicates a part of the answer that is incorrect.
Highlighter	To draw attention to a particular aspect or to indicate where parts of an answer have been combined.
	Too vague.

REP	Repetition
NE	No examples or not enough.

Annotation	Meaning
BOD	Benefit of Doubt.
NAQ	Not Answered Question.
SEEN	Indicates that work or a page has been seen including blank answer spaces and blank pages.
FT	Follow through.
I	Ignore

6(a)	1 **mark** for each **set** of lines from each denary number **Denary number** / **Equivalent unsigned value**  	**4**

Total: 75 marks

6(b)	**1 mark** for working **1 mark** for answer. 1101 0100 1111 1111 1100 1111 1 1101 0001 1110 1111 1 1 111	**2**	No marks for answer if no evidence of using binary addition.
6(c)	**1 mark** for correct answer 1713	**1**	
6(d)	**1 mark** for correct answer 1000000000000000	**1**	Last answer

Question 7 · Oct/Nov 2023 Paper 12, original Q6 (a),(b),(c) [8 marks]

Question	Answer	Marks

7(a)	**1 mark** for each bullet point (**max 3**) Lossy compression (ticked) • Loss of quality will not be noticed • Needs to be viewed in real time so less bandwidth needed if file size smaller • Smaller file sizes will reduce buffering so the video will play more smoothly • Viewers may watch on different devices, so may not need high quality resolution Lossless compression (ticked) • Original recording may not have been made in high resolution • Could be streaming to high bandwidth devices • The reduction in the file size is sufficient for the receiving device • Viewers do not want any loss of quality	**3**

7(b)	**1 mark** for each bullet point (**max 3**) Increase sampling resolution • … the number of bits used for each sample is increased • … there will be more values available to represent each sample // more amplitudes can be represented • ... each binary amplitude/note in the digital recording is closer to the analogue amplitude/note • … quantisation errors are reduced • … the digital soundwave is closer to the original analogue soundwave Decrease sampling resolution • … the number of bits used for each sample is decreased • … there will be fewer values available to represent each sample // fewer amplitudes can be stored • ... each binary amplitude/note in the digital recording is further from the analogue amplitude/note • … quantisation errors are increased • … the digital soundwave is less like the original analogue soundwave	**3**
7(c)	**1 mark** for answer, **1 mark** for working Answer: 2.5 mebibytes Working: (2048 × 1024 × 10) / (8 × 1024 × 1024)	**2**

Question 8 · Oct/Nov 2024 Paper 13, original Q6 (a),(b),(c) [10 marks]

Question	Answer	Marks
8(a)	**1 mark** for identification **1 mark** for matching description e.g. • ASCII • 7/8 bits per character // represents 128/256 characters // represents all characters from Latin alphabet • UNICODE • 8/16/32 bits per character // represents 256/65536+ characters // represents all characters in all languages	**2**
8(b)(i)	**1 mark** for: 256 // 2^8	**1**
8(b)(ii)	**1 mark** for: Increased file size	**1**
8(b)(iii)	**1 mark** for each bullet point (**max 2**) e.g. • The change may not be noticeable // Data removed is usually not noticed by the human eye • … for example, changes in shade/detail • It produces a larger decrease in file size compared to lossless // Lossy decreases file size considerably	**2**

8(c)(i)	**1 mark** for each bullet point (**max 2**) • Value/magnitude/size of the analogue sound wave is measured a set number of times each second/time / at set intervals • Each sample/reading/measurement is given the binary number and stored in sequence	**2**
8(c)(ii)	**1 mark** for each correct point and **1 mark** for matching expansion e.g. • Decrease sample rate • … fewer samples/readings/measurements stored per second // fewer bits per second stored • Decrease sample resolution • … fewer bits per sample/reading/measurement // each sample has fewer bits • Sound outside of set/human hearing range is removed • … fewer measurements are stored / decreases the number of possible binary values so fewer bits are stored	**2**

Question 9 · Oct/Nov 2024 Paper 13, original Q8 (b) [3 marks]

Question	Answer	Marks

9(a)	**1 mark** for each bullet point for the method (**max 2**) e.g. • Flip each bit then add 1 … • … method of converting the new binary number into denary • Most significant 1 bit is treated as the corresponding negative denary value … • … add the other positive corresponding denary values **1 mark** for correct conversion Denary value: −97	**3**